A Simscape Language Thyristor Model with Reverse Recovery

Lumped-Charge (Lauritzen–Ma) Dynamics with Latching Logic and a Physical t_q Constraint

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Abstract

Simscape Electrical does not provide a thyristor block with junction charge dynamics: the *Thyristor (Piecewise Linear)* block models reverse recovery only as a thermal loss term (from R2023a), and the Specialized Power Systems *Detailed Thyristor* models only the circuit-commutated turn-off time t_q without the reverse recovery current waveform. This document describes a custom Simscape language component that combines the piecewise-linear SCR static characteristic with the Lauritzen–Ma lumped-charge model, so that the terminal current reproduces the reverse recovery peak I_{rr}, the stored charge Q_{rr} and the recovery tail. The latching and holding behaviour is implemented with an event variable, and the turn-off time constraint t_q is reproduced physically through a residual-charge re-trigger condition complemented by an explicit timer for the late recombination phase that the two-charge model cannot represent. Parameter extraction from datasheet quantities is discussed and carried out in full for the ABB 5STP 26N6500 6.5 kV/2880 A phase-control thyristor. The model is then validated in a six-pulse line-commutated rectifier, where the recovered charge agrees with the datasheet within 5 %; the structural limits of the two-charge formulation are documented, and the relevant literature is referenced.

1 Motivation

The thyristor is a bipolar four-layer (pnpn) device. During conduction the junctions are flooded with minority carriers; at natural commutation the anode current does not stop at zero but reverses while the stored charge is extracted, producing a reverse recovery transient characterized by the peak current I_{rr}, the recovered charge Q_{rr} and the recovery time t_{rr}. Phase-controlled thyristors, designed with long carrier lifetime for low on-state voltage, store comparatively large charge, so the recovery transient is significant. Its practical consequences include:

- overvoltage spikes $L\,\mathrm{d}i/\mathrm{d}t$ on the commutation inductance during current snap-off, which drive RC snubber and clamp design;
- switching (recovery) losses;
- the circuit-commutated turn-off time t_q: after recovery, residual charge in the n and p bases must recombine before forward voltage can be reapplied, otherwise the device re-fires spontaneously. t_q is typically much longer than t_{rr} (50 µs to 200 µs for phase-controlled devices, 10 µs to 30 µs for fast thyristors) and sets the extinction margin (angle γ) in line-commutated converters.

An electrical (waveform-level) reverse recovery model is therefore required whenever snubber stress, commutation overvoltage or recovery-driven interactions must be simulated. Since no standard block provides it, the component below implements it in the Simscape language.

2 Theoretical background: the lumped-charge model

2.1 Lauritzen–Ma two-charge model

Lauritzen and Ma [1] describe the excess minority-carrier charge in the lightly doped base with two lumped state quantities: the charge q_E injected at the junction boundary, which follows the static device characteristic instantaneously, and the charge q_M stored in the middle of the base region, which follows q_E with finite dynamics. Charge is exchanged between the two nodes by diffusion with a cross-over time constant T_M, and q_M recombines with the carrier lifetime τ. The model equations are:

$$i(t) = \frac{q_E - q_M}{T_M}, \tag{1}$$

$$\frac{\mathrm{d}q_M}{\mathrm{d}t} = \frac{q_E - q_M}{T_M} - \frac{q_M}{\tau}. \tag{2}$$

The injected charge is tied to the static characteristic $i_s(v)$ through

$$q_E = (\tau + T_M)\, i_s(v), \tag{3}$$

where the gain $(\tau + T_M)$ is chosen so that in forward steady state ($\mathrm{d}q_M/\mathrm{d}t = 0$) the terminal current reduces exactly to $i = i_s(v)$. This is the same formulation documented for the charge dynamics option of the Simscape *Diode* block.

During commutation, the external circuit forces i_s (and hence q_E) down and negative, while q_M can only decrease at the rate allowed by (2). The difference $q_E - q_M$ in (1) produces the negative recovery current. Once the junction blocks ($q_E \approx 0$), the remaining q_M decays with the effective recovery time constant

$$\tau_{rr} = \frac{\tau\, T_M}{\tau + T_M}, \tag{4}$$

which gives the model its characteristic soft exponential recovery tail. The snappiness of the recovery is controlled by the ratio T_M/τ: small T_M/τ gives abrupt (snappy) recovery, larger values give soft recovery.

The 1991 model was subsequently extended to include forward recovery and emitter recombination [2], generalized into the systematic lumped-charge modeling technique with parameter extraction from datasheets [3], and applied to multi-layer devices including SCRs and GTOs [4]. A practical implementation of the diode and thyristor models with automatic parameter extraction was given by Courtay for the Saber/MAST environment [5], which is the reference cited by the MathWorks documentation for the Diode charge model. The device physics background (charge control, base charge storage, origin of t_q) is covered by Baliga [7].

2.2 Application to the thyristor

A rigorous lumped-charge thyristor model represents the pnpn structure as two coupled transistors with separate base charge nodes [4]. For system and converter level simulation a simpler behavioural structure is adequate: the static characteristic is the piecewise-linear SCR model with a latch,

$$i_s = \begin{cases} \dfrac{v - V_f\,(1 - R_{on}G_{off})}{R_{on}}, & \text{device on,} \\ v\, G_{off}, & \text{device off,} \end{cases} \tag{5}$$

identical to the *Thyristor (Piecewise Linear)* block, and the charge dynamics (1)–(3) are wrapped around it. The latch state evolves by discrete events:

- **turn-on**: gate signal above the trigger threshold V_{gt} while $v > 0$;
- **turn-off**: anode current below the holding current I_h. The gate state must *not* enter this condition: a real thyristor commutates naturally even with the gate still high, and gating the turn-off on a low gate signal delays the recorded turn-off instant whenever the firing pulse is a long block;
- t_q **re-trigger**: if forward voltage above a threshold V_{th} is reapplied while the *excess* charge exceeds Q_{th}, or within a time t_q of the turn-off event, the device turns back on spontaneously.

Excess charge, not stored charge

The re-trigger condition must be written on the excess charge

$$q_{exc} = q_M - \tau\, G_{off}\, v, \tag{6}$$

not on q_M itself. In the blocking state q_M does not decay to zero: setting $\mathrm{d}q_M/\mathrm{d}t = 0$ in (2) with $q_E = (\tau + T_M)\, G_{off}\, v$ gives the leakage steady state $q_{M,\infty} = \tau\, G_{off}\, v$. For the device of Section 4.1 with $\tau = 100\,\mu\mathrm{s}$ and the worst-case $G_{off} = 1 \times 10^{-4}\,\Omega^{-1}$ this is $4.3\,\mu\mathrm{C}$ at V_{DWM} and $6.5\,\mu\mathrm{C}$ at V_{DRM} — far above any sensible Q_{th}. A condition written on q_M therefore fires the device as soon as forward voltage returns, mistaking stationary leakage for residual recovery charge. The combination (6) vanishes in any steady state, forward or reverse, and equals the residual charge during turn-off. A finite V_{th} (50 V to 100 V) additionally prevents ringing around zero from triggering the condition.

Threshold and timer

In the blocking state $q_E \approx 0$, so (2) reduces to $\mathrm{d}q_M/\mathrm{d}t = -q_M(1/T_M + 1/\tau) = -q_M/\tau_{rr}$: the residual charge decays with the *recovery* time constant τ_{rr}, not with the lifetime τ. With $q_M(0) = a\,\tau^2$ at the current zero crossing (Section 4.1), the threshold that realizes a given turn-off time is

$$Q_{th} = a\,\tau^2\, e^{-t_q/\tau_{rr}}. \tag{7}$$

For the parameter set of Section 4.1 ($\tau_{rr} = 65.5\,\mu\mathrm{s}$, $a = 1.7\,\mathrm{A\,\mu s^{-1}}$) this gives $Q_{th} \approx 0.09\,\mu\mathrm{C}$ for the datasheet $t_q = 800\,\mu\mathrm{s}$: five decades below $q_M(0)$, but entirely representable in double precision *once the leakage baseline has been removed* by (6). The charge mechanism alone is therefore sufficient, and the explicit timer carried by the component (the turn-off event stores its time in an event variable; reapplied forward voltage within t_q of that instant re-fires the device) is a redundant safeguard rather than a necessity. A more physical alternative to both is a third, slow charge node in the spirit of the multi-node lumped-charge technique [3], fed by q_M and governing only the re-trigger condition.

3 Simscape language implementation

The component uses electrical anode and cathode nodes and a physical-signal gate input (replace with an electrical gate node plus R_g if gate-circuit loading matters). The latch is an integer event variable updated in the `events` section (Simscape language R2019b or later). The full source is listed below.

Listing 1: `thyristor_rr.ssc`

```
component thyristor_rr
% Thyristor with Reverse Recovery (Lauritzen-Ma charge dynamics)
```

```
% ------------------------------------------------------------
% Piecewise-linear SCR static characteristic + Lauritzen-Ma lumped
% charge model for reverse recovery, plus latching/holding logic.
% Defaults: ABB 5STP 26N6500 (6.5 kV / 2880 A phase-control thyristor),
% typical values at Tvj = 125 degC, validated in a six-pulse bridge.
%
% Static PWL characteristic (as in Thyristor (Piecewise Linear)):
% on: i_s = (v - Vf*(1 - Ron*Goff))/Ron
% off: i_s = v*Goff
%
% Charge model (as documented for the Simscape Diode block):
% qE = (tau + Tm) * i_s injected charge
% der(qM) = (qE - qM)/Tm - qM/tau stored charge
% i = (qE - qM)/Tm terminal current
%
% tau_rr = tau*Tm/(tau+Tm) is the recovery tail time constant.
%
% Latching logic (event variable 'on'):
% turn ON : gate signal G > Vgt while v > 0
% turn OFF : i < Ih (gate state is IRRELEVANT: a real thyristor
% commutates naturally even with the gate still high)
% re-trigger: forward voltage above Vth while EITHER
% (a) EXCESS charge qExc > Qth, or
% (b) time - toff < tq (explicit timer, belt and braces)
%
% IMPORTANT - excess charge:
% In the blocking state qM does NOT decay to zero: it settles at the
% leakage steady state qM_inf = tau*Goff*v. Comparing qM itself
% against Qth therefore makes the device self-trigger as soon as
% forward voltage returns (for the 26N6500 at VDRM, tau*Goff*v is
% several microcoulombs). The re-trigger condition must use the
% excess charge qExc = qM - tau*Goff*v, which is zero in any steady
% state and equals the residual recovery charge during turn-off.
%
% Parameter sets (see accompanying document):
% A (default, validated): tau = 100 us, Tm = 190 us
% -> Qrr within 5 % of datasheet, tail 66 us, IRM ~20 % low
% B (IRM-centred alternative): tau = 85 us, Tm = 100 us
% -> IRM and Qrr both centred, tail shortens to ~46 us
% The two-charge model has only two degrees of freedom and cannot
% match IRM, Qrr and the tail constant simultaneously.

  nodes
    a = foundation.electrical.electrical; % A:left
    k = foundation.electrical.electrical; % K:right
  end

  inputs
    G = {0, 'V'}; % G:left gate command (compare against Vgt)
  end

  parameters
    Vf = {1.12, 'V' }; % Forward voltage (VT0, 125 degC)
    Ron = {0.29e-3, 'Ohm' }; % On-state resistance (rT, 125 degC)
    Goff = {1e-5, '1/Ohm'}; % Off-state conductance (IRRM/VRRM, worst case)
    Vgt = {1.2, 'V' }; % Gate trigger threshold
    Vth = {50, 'V' }; % Forward voltage threshold for re-trigger
    Il = {0.25, 'A' }; % Latching current (125 degC)
```

```
  Ih = {0.075, 'A' }; % Holding current (125 degC)
  tau = {100e-6, 's' }; % Carrier lifetime (transit time)
  Tm = {190e-6, 's' }; % Charge cross-over time constant
  Qth = {1e-6, 'C' };  % Excess-charge re-trigger threshold
  tq = {800e-6, 's' }; % Circuit-commutated turn-off time (timer)
end

parameters (Access = private)
  Kq = tau + Tm; % qE gain so that i -> i_s in steady state
end

variables
  i = {0, 'A'}; % Through variable A->K
  v = {0, 'V'}; % Across variable A-K
  qM = {value = {0, 'C'}, priority = priority.high}; % Stored charge
end

variables (Event = true)
  on = 0; % Latch state: 0 = blocking, 1 = conducting
  toff = {-1, 's'}; % Time of last turn-off event
end

intermediates
  % Static PWL current
  i_s = if on > 0.5, (v - Vf*(1 - Ron*Goff))/Ron else v*Goff end;
  % Injected charge
  qE = Kq * i_s;
  % Excess charge over the leakage steady state (see header)
  qExc = qM - tau*Goff*v;
end

branches
  i : a.i -> k.i;
end

equations
  v == a.v - k.v;
  % Lauritzen-Ma lumped charge dynamics
  der(qM) == (qE - qM)/Tm - qM/tau;
  i == (qE - qM)/Tm;
end

events
  when edge((G > Vgt) && (v > {0,'V'}) && (on < 0.5))
    on = 1; % gate turn-on
  elsewhen edge((v > Vth) && (on < 0.5) && ...
              ((qExc > Qth) || (time - toff < tq)))
    on = 1; % tq violation: spontaneous re-fire
  elsewhen edge((i < Ih) && (on > 0.5))
    on = 0; % natural commutation below holding
    toff = time; % start tq timer
  end
end

annotations
  a : Side = left;
  k : Side = right;
  G : Side = bottom;
```

```
  end

end
```

Implementation notes:

- The stored charge q_M is declared with high initialization priority and zero initial value (device initially blocking and discharged).
- The static current i_s and injected charge q_E are `intermediates`, so the DAE keeps only q_M as a continuous state.
- The system is stiff (τ, T_M in the microsecond range against milliohm on-resistance): use the `daessc` solver or `ode23t` with tight tolerances.
- Always include the commutation (source) inductance in the test circuit; without it the current slope is unbounded and the recovery transient degenerates.
- The turn-off event stores the event time in the real-valued event variable `toff`; the t_q timer condition compares `time - toff` against the parameter `tq`.
- The latching current I_l is declared but the present event logic latches directly on gate turn-on. Strict latching behaviour (turn-off if the gate is removed before $i > I_l$) requires one additional event flag and transition.

4 Parameter extraction from datasheet data

Datasheets for phase-controlled thyristors specify Q_{rr}, I_{RM} and t_q at a reference operating point, and Q_{rr}, I_{RM} as functions of the commutation slope $a = |\mathrm{d}i/\mathrm{d}t|$. A practical extraction sequence:

1. τ_{rr} **from the reference point.** Decompose the recovered charge into the triangle up to the peak (duration I_{RM}/a) plus the exponential tail (charge $I_{RM}\tau_{rr}$):

$$Q_{rr} = \frac{I_{RM}^2}{2a} + I_{RM}\,\tau_{rr} \quad \Longrightarrow \quad \tau_{rr} = \frac{Q_{rr}}{I_{RM}} - \frac{I_{RM}}{2a}. \tag{8}$$

2. **Split** τ/T_M**.** Equation (8) fixes only the parallel combination. The split is constrained by the datasheet curves $Q_{rr}(a)$ and $I_{RM}(a)$: the growth of Q_{rr} with a depends mainly on τ (the charge available at the current zero crossing), the peak shape on T_M. Choose a trial τ (lifetime of the device class), set $T_M = \tau\,\tau_{rr}/(\tau - \tau_{rr})$, and iterate on τ against the curves.

3. **Validation sweep**: simulate a controlled-$\mathrm{d}i/\mathrm{d}t$ commutation test harness (single pulse with series inductance), sweep a over the datasheet range, extract I_{RM} and Q_{rr} from the waveforms, and compare against the datasheet curves. A power-law fit $I_{RM}(a)$, $Q_{rr}(a)$ of the datasheet points is convenient both for the comparison and for extrapolation beyond the specified range.

4. t_q **and** Q_{th}: set the timer parameter t_q directly from the datasheet. Q_{th} only shapes the early-phase charge re-trigger and is uncritical; (7) gives the turn-off time it realizes on its own.

5. **Temperature**: τ, and hence Q_{rr} and t_q, increase with junction temperature. For a fixed-temperature study parameterize at the worst case (maximum T_j); a temperature-dependent model requires making τ a function of a thermal port variable.

The extraction methodology follows [3] and [5].

4.1 Worked example: ABB 5STP 26N6500

The ABB 5STP 26N6500 [6] is a 6.5 kV / 2880 A phase-control press-pack thyristor. The datasheet recovery data (all at $T_{vj} = 125\,°\mathrm{C}$, $I_T = 2000\,\mathrm{A}$, $V_R = 200\,\mathrm{V}$, $\mathrm{d}i_T/\mathrm{d}t = -1.5\,\mathrm{A\,\mu s^{-1}}$) are: $Q_{rr} = 3500/4940/5800$ µC (min/typ/max), $I_{RM} = 60/77/90$ A, $t_q \leq 800\,\mu\mathrm{s}$.

Recovery time constant. Applying (8) at $a = 1.5\,\mathrm{A\,\mu s^{-1}}$:

	min	typ	max
Q_{rr}/I_{RM} [µs]	58.3	64.2	64.4
$I_{RM}/2a$ [µs]	20.0	25.7	30.0
τ_{rr} [µs]	38.3	38.5	34.4

The three corner values agree within 12 %, so $\tau_{rr} \approx 38.5\,\mu\mathrm{s}$ (typical) is well constrained. Consistency check: time to peak $I_{RM}/a = 51.3\,\mu\mathrm{s}$, triangle charge 2.0 mC plus tail charge $I_{RM}\tau_{rr} = 2.9\,\mathrm{mC}$ reproduce $Q_{rr} = 4.9\,\mathrm{mC}$.

Lifetime split. A first guess taken from device physics — the lifetime of a 6.5 kV phase-control device, several hundred microseconds — is *wrong* for this model, and the reason is worth stating explicitly. During a slow current ramp $i = I_0 - at$ the charge equation has the quasi-stationary solution $q_M(t) = \tau\, i(t) + a\tau^2$, so the charge left at the current zero crossing is

$$q_M(0) = a\,\tau^2. \tag{9}$$

With $\tau = 300\,\mu\mathrm{s}$ and $a = 1.5\,\mathrm{A\,\mu s^{-1}}$ this gives 135 mC, 27 times the datasheet Q_{rr}, and a reverse peak of several kiloamperes: the transient is violent enough to make the solver fail. In this model τ is therefore *not* the physical deep-base lifetime but an effective parameter tied to the extractable charge, $\tau \approx \sqrt{q_M(0)/a}$ with $q_M(0)$ of the order of Q_{rr}.

Combining (9) with the peak and charge relations $I_{RM} \approx a\tau^2/T_M$ and $Q_{rr} \approx a\tau^3/(\tau + T_M)$ and solving at the datasheet reference point gives

$$\tau \approx 100\,\mu\mathrm{s}, \qquad T_M \approx 190\,\mu\mathrm{s}, \tag{10}$$

which reproduces $I_{RM} = 77\,\mathrm{A}$ and $Q_{rr} = 4.9\,\mathrm{mC}$ at $1.5\,\mathrm{A\,\mu s^{-1}}$, with $\tau_{rr} = 65.5\,\mu\mathrm{s}$. Note that $T_M > \tau$ is legitimate: it merely lowers the gain $(\tau + T_M)/T_M$ and makes the model numerically better behaved. The split should still be checked against the datasheet curves $Q_{rr}(a)$ and $I_{RM}(a)$ (Figs. 8–9 of [6], $a = 1$ to $30\,\mathrm{A\,\mu s^{-1}}$) with the validation of step 3.

Static and latch parameters. On-state: $V_f = V_{T0} = 1.12\,\mathrm{V}$, $R_{on} = r_T = 0.29\,\mathrm{m\Omega}$ (125 °C, valid 1300 A to 4000 A; for accuracy over the full current range the datasheet provides the four-term model $V_T = A + BI_T + C\ln(I_T{+}1) + D\sqrt{I_T}$, which can replace the PWL expression in the `i_s` intermediate). Off-state conductance from the leakage specification $I_{RRM} \leq 600\,\mathrm{mA}$ at 6.5 kV, 125 °C: $G_{off} \approx 1 \times 10^{-4}\,\Omega^{-1}$ (worst case). Latch: $I_h = 75\,\mathrm{mA}$, $I_l = 250\,\mathrm{mA}$ (125 °C). Gate thresholds $V_{GT} \leq 2.6\,\mathrm{V}$, $I_{GT} \leq 400\,\mathrm{mA}$ are specified at 25 °C and decrease with temperature; for the signal comparator of the model $V_{gt} = 1.2\,\mathrm{V}$ is adequate.

Turn-off time. From (7) with $\tau_{rr} = 65.5\,\mu\mathrm{s}$ and $q_M(0) = a\tau^2 = 15\,\mathrm{mC}$ at the reference slope, the datasheet $t_q = 800\,\mu\mathrm{s}$ corresponds to $Q_{th} \approx 0.09\,\mu\mathrm{C}$; 0.1 µC is used. This is workable only with the excess-charge formulation (6): compared against q_M itself, the leakage steady state $\tau G_{off} v = 6.5\,\mu\mathrm{C}$ at V_{DRM} would exceed the threshold by two orders of magnitude and fire the device on every return of forward voltage. The timer parameter is set to the datasheet value as a redundant safeguard.

Resulting configuration. The complete parameter set (typical values, $T_{vj} = 125\,^{\circ}\mathrm{C}$), used as defaults in the component listing above:

Parameter	Value	Unit	Source
V_f	1.12	V	V_{T0}
R_{on}	0.29	mΩ	r_T
G_{off}	10^{-4}	Ω^{-1}	I_{RRM}/V_{RRM}
V_{gt}	1.2	V	V_{GT}, derated
V_{th}	50	V	anti-chatter, re-trigger
I_l	0.25	A	I_L, 125 °C
I_h	0.075	A	I_H, 125 °C
τ	100	µs	fit on I_{RM}, Q_{rr}
T_M	190	µs	fit on I_{RM}, Q_{rr}
Q_{th}	0.1	µC	from (7)
t_q	800	µs	datasheet maximum

5 Validation in a six-pulse bridge

The component was validated in a six-pulse line-commutated rectifier with the parameter set above, an RC snubber of 22 Ω / 1 µF across each device (the values at which the datasheet turn-off energy curves are measured), and short gate pulses. Conducted current 764 A, commutation slope $1.71\,\mathrm{A\,\mu s^{-1}}$, solver step 0.5 µs.

5.1 Effect of the snubber

Without the snubber the anode voltage reached 21.9 kV, i.e. $3.4\,V_{DRM}$, and the current stepped from zero to I_{RM} within a single solver step: the triangular phase of the recovery was absent, since the latch opens the PWL characteristic exactly at the current zero and the stored charge discharges abruptly through q_M/T_M. With the snubber in place the voltage peak falls to 1.14 kV, the maximum $\mathrm{d}v/\mathrm{d}t$ to $1140\,\mathrm{V\,\mu s^{-1}}$ (against the specified $\mathrm{d}v/\mathrm{d}t_{crit} = 3000\,\mathrm{V\,\mu s^{-1}}$), and the rise from the zero crossing to I_{RM} lengthens to 28.5 µs. The triangular phase is thus recovered by the circuit: holding the voltage prevents the snap, and the external loop keeps imposing the ramp $-a$. The apparent step of the unsnubbed case was largely a circuit artefact rather than a model defect.

5.2 Separating the snubber current

If the measured current is taken on the branch rather than on the device, it includes the snubber displacement current, which is not negligible: notches in v_{ak} produced by the commutations of the other device pairs generate current spikes whose peak is set by $\Delta v/R_s$ (measured 7.8 A against a predicted 8.0 A for $\Delta v = 176$ V, and 26.4 A against 29 A for $\Delta v = 640$ V). These spikes are easily mistaken for spurious re-triggering; they are not, since a genuine re-fire would draw the full link current. The device current is recovered by integrating

$$\frac{\mathrm{d}i_{sn}}{\mathrm{d}t} = \frac{1}{R_s}\left(\frac{\mathrm{d}v}{\mathrm{d}t} - \frac{i_{sn}}{C_s}\right), \qquad i_{dev} = i - i_{sn}. \tag{11}$$

5.3 Measured recovery

	branch	device	datasheet (typ.)
I_{RM} [A]	73.4	65.3	83
Q_{rr} [mC]	6.07	5.56	5.30
tail constant [µs]	54.0	66.0	—

The datasheet column is the reference point scaled to $a = 1.71\,\mathrm{A\,\mu s^{-1}}$ with the power law of Figs. 8–9. The recovered charge agrees within 5 % and the measured tail constant matches the predicted $\tau_{rr} = \tau T_M/(\tau + T_M) = 65.5\,\mu\mathrm{s}$ to better than 1 %, confirming the internal consistency of the implementation. The peak current is low by about 20 %.

Centring I_{RM} as well would require $\tau \approx 85\,\mu\mathrm{s}$, $T_M \approx 100\,\mu\mathrm{s}$, at the cost of shortening the tail constant to 46 µs. This is not a defect of the fit but a structural property: the two-charge model has *two* degrees of freedom and the recovery transient has three observable features (I_{RM}, Q_{rr}, tail constant). Since the datasheet specifies only the first two, the choice of which pair to match should follow the intended use — charge and tail for snubber energy and commutation margin, peak current for reverse voltage stress.

6 Limitations

The model is behavioural at the latch level: it does not represent the two coupled transistor charges of the pnpn structure, so it does not capture gate-controlled turn-off (GTO operation), $\mathrm{d}v/\mathrm{d}t$-induced triggering through junction capacitance (the 5STP 26N6500 specifies $\mathrm{d}v/\mathrm{d}t_{crit} = 3000\,\mathrm{V\,\mu s^{-1}}$, not enforced by the model), or the plasma spreading phase at turn-on ($\mathrm{d}i/\mathrm{d}t$ limitation). Junction capacitance in the blocking state is not included and can be added in parallel externally or as an additional charge term. With two free parameters it cannot match I_{RM}, Q_{rr} and the recovery tail constant simultaneously (Section 5.3), and the lifetime τ is an effective fitting parameter rather than the physical carrier lifetime of the device (Section 4.1). Within its scope — natural commutation, reverse recovery waveforms, snubber and clamp stress, t_q margin studies in line-commutated converters — the two-charge model is well established and adequate.

References

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